

Christof Wellan

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Experience

Robotics Software Engineering Intern, Avientus (ETH Spin-off) September 2025 – February 2026

- Led a real-time visual detection/tracking and semantic target-search pipeline for last-mile drone delivery in a 3-person team (SPRIND Fully Autonomous Flight Challenge 2.0; advanced to stage 2, €325k award), including model fine-tuning and Jetson AGX Orin deployment. Integrated it into a ROS 2 autonomy stack and supported visual-inertial SLAM and path-planning integration.

Summer Intern, BOSCH GmbH July 2023 – August 2023

- Performed Hardware-in-the-Loop (HIL) tests on inverter units by deliberately triggering fault flags to validate error-handling procedures; documented test cases and outcomes to ensure compliance with technical specifications.

Teaching Assistant (Mechanics of Solid Bodies), TU Vienna October 2023 – June 2024

Summer Intern, HOERBIGER Wien GmbH July 2022 – August 2022

Projects

SO-ARM101 Robot Learning (ACT Imitation Learning) – Independent Project March 2026 – Present

- Trained a real-world SO-ARM101 visuomotor pick-and-place policy, comparing ACT, diffusion policies, a VLA model, and DAGger; built an Isaac Lab RL pipeline with privileged-teacher/student distillation and ongoing sim-to-real transfer; fine-tuned a Nano World Model on public and self-collected robot data.

Master Thesis – ETH Product and Development Group March 2026 – September 2026

- Developing task-aware grasp planning and execution for autonomous dual-arm assembly on KUKA iiwa 7 robots: generating part-specific grasps offline, filtering them for assembly feasibility via constraint satisfaction, and selecting/executing feasible grasps online from estimated object poses with an RL-based real-robot controller.

EPFL Robohack 2026 – Unitree G1 Robot Control Interface – EPFL AI Team May 2026

- Won 2nd place by building a control interface for the Unitree G1, enabling users to run predefined motions, generate new motions from text prompts with Kimodo-G1, preview them in MuJoCo, and deploy approved motions on the real robot. Also built a Meta Quest teleoperation pipeline and collected demonstration data for future imitation learning.

Semester Project – ETH Robotic Systems Lab March 2025 – June 2025

- Built a vision-language model (VLM) agent for autonomous high-level excavator planning in Isaac Lab, enabling a simulated excavator to clear a rock field. Integrated ROS 2 action servers to execute low-level control policies, resulting in a successful demonstration.

Education

ETH Zürich – MSc in Mechanical Engineering (focus on Robotics) September 2024 – September 2026

- Coursework included: control (e.g. MPC, RL), perception, estimation, machine learning, and embedded systems

TU Vienna – BSc in Mechanical Engineering September 2021 – July 2024

- Passed with distinction
- Coursework included: FEM, Multibody Dynamics, Modal Analysis, Numerical Methods

Skills

Programming: Python (e.g. PyTorch), MATLAB/Simulink, C/C++, Isaac Lab, MuJoCo, ROS 2, Docker, Git

CAD: SolidWorks, Creo, Autodesk Fusion

Achievements

Merit Scholarship – TU Vienna 2022 and 2024

Languages

English – Level C2

German – Native